

Dynamic ETA Calculation in the Waymo Dataset

Module Overview

This module computes a **vehicle's Estimated Time of Arrival (ETA)** to its relevant stop line. The approach is a short-horizon, kinematic calculation: it first identifies the lane a vehicle is in and its direction of travel to determine the correct stop line. It then computes the distance along that lane's path to the stop line and divides this by the vehicle's current speed. The resulting ETA is clipped to a reasonable range (0.1s to 8.0s) to avoid outliers that can arise from the dataset's 10Hz sampling rate and short 8-second future window.

The Waymo Open Motion Dataset Context

The script operates on the Waymo Open Motion Dataset (WOMD), which provides a highly structured and pre-processed environment for traffic analysis. Unlike raw sensor data, the dataset consists of discrete 20-second scenarios packaged in the `.tfrecord` format. For the purposes of this algorithm, the script utilizes two key components of the dataset:

- **High-Fidelity Map Data:** Each scenario includes a detailed map with features represented as polylines. The script specifically uses **lane centerlines** to associate vehicles with their path of travel. The start and end points of these lane polylines serve as potential stop lines.
- **Tracked Object States:** The dataset contains data for all road users, which have already been classified (e.g., as vehicles) and tracked over time. The script leverages the per-timestep state of each vehicle, which provides its precise **position** (centerx,centery), **velocity vector** (velocityx,velocityy), and **heading**.

This pre-processed data allows the algorithm to bypass complex object detection and tracking, focusing directly on the kinematics and lane-level behavior of each vehicle.

The ETA Calculation Algorithm

The core of the module is a multi-step process that is executed for each vehicle at every 0.1-second timestep in a scenario. The goal is to derive a reliable, short-term ETA to the next stop line. The calculation is only performed for vehicles exceeding a minimum speed of **0.5 m/s** to filter out parked or stationary vehicles.

1. Lane Assignment

First, a slow-moving vehicle must be accurately associated with a specific lane. The algorithm iterates through all lane centerlines provided in the map data. For each lane, it calculates the shortest (perpendicular) distance from the vehicle's current position to that lane's polyline. The vehicle is provisionally assigned to the lane with the minimum perpendicular distance.

To ensure accuracy and exclude vehicles that are not actively in a lane (e.g., those in a parking lot or on the shoulder), an assignment threshold is used. If the vehicle's distance to the closest lane centerline is greater than the `LANE_ASSIGNMENT_THRESHOLD` of **5.0 meters**, the assignment is discarded, and no ETA is calculated for that vehicle at that timestep.

2. Dynamic Stop Line Determination

As there are no pre-defined stop lines in waymo's open motion dataset, we are dynamically setting stop lines for each vehicle in a lane.

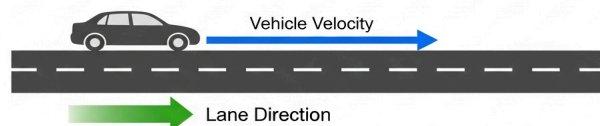
A critical challenge is determining which end of the lane's polyline represents the upcoming stop line relative to the vehicle's direction of travel. The algorithm solves this dynamically using vector mathematics:

- It identifies the specific segment of the lane polyline that the vehicle is currently on.
- It calculates the **dot product** of the vehicle's velocity vector and the direction vector of that lane segment.
 - If the dot product is **positive or zero**, the vehicle is moving in the same direction as the polyline's vertex order. Therefore, the stop line is defined as the **last point** of the polyline.
 - If the dot product is **negative**, the vehicle is moving against the polyline's vertex order. In this case, the stop line is defined as the **first point** of the polyline.

This method robustly determines the correct stop line regardless of how the lane was defined in the map data, making the system adaptable to lanes with different orientations and directions.

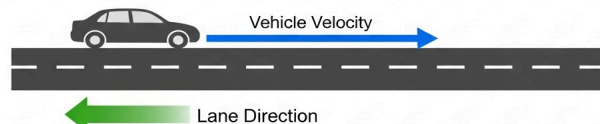
Dynamic Stop Line Calculation —>

Positive Dot Product (Vectors Aligned)



Result: Dot Product > 0 , Stop Line is the **END** of the lane.

Negative Dot Product (Vectors Opposed)



Result: Dot Product < 0 , Stop Line is at the **START** of the lane.

3. Path Distance and ETA Calculation

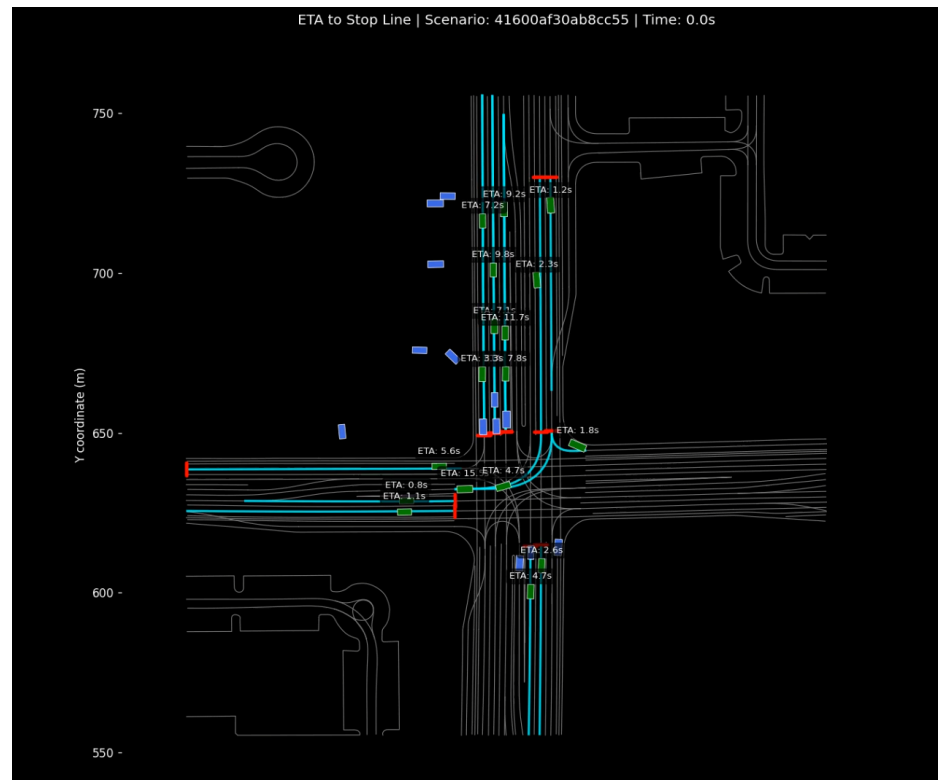
With the vehicle assigned to a lane and the correct stop line identified, the algorithm calculates the remaining distance. This is not a straight-line (Euclidean) distance but the **path distance** along the lane's centerline. It is computed by summing the length of all remaining polyline segments from the vehicle's current projected position on the lane to the dynamically determined stop line.

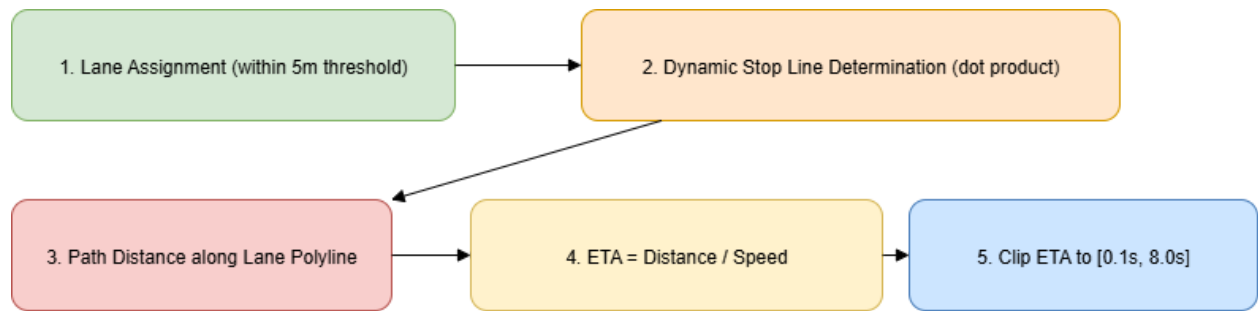
Finally, the kinematic ETA is calculated with a simple formula:

$$\text{ETA} = \text{Current Speed} / \text{Path Distance to Stop Line}$$

To improve stability, the final ETA value is clipped to a range of **0.1 to 8.0 seconds**. This prevents extreme values caused by instantaneous speed fluctuations or situations where a vehicle is very far from a stop line, which falls outside the scope of a short-horizon prediction. The final output of the script is a video visualization that plots each vehicle, highlights its assigned lane, marks the identified stop line, and displays the calculated ETA in real-time.

An example of the module working on a scenario





Working pipeline diagram of the algorithm

Sources

Dataset : Waymo's Open Motion Dataset - [Waymo's Open Motion Dataset](#)